Ranked reading list for the base-operator program

A self-contained extraction guide built from the literature memo

Purpose

This document turns the larger literature memo into a practical reading sequence. The ranking is not a judgment of mathematical importance in general. It is a judgment of *expected payoff for the present research program*, whose central objects are the base operator

$$L_\lambda \quad \text{and} \quad A_\lambda = L_\lambda + C_\lambda I + K_\lambda,$$

with emphasis on the following goals:

1. simplify and strengthen the current paper;
2. make the architecture portable to Dedekind zeta, automorphic, and Rankin-Selberg settings;
3. derive stronger spectral conclusions and sharper inequalities;
4. extract new structural properties of L_λ;
5. plan realistic routes toward simple-even for A_λ;
6. identify productive interaction with the CCM / *Zeta Spectral Triples* program;
7. strengthen the number-theoretic side of the operator analysis.

A reference is ranked highly when it is likely to shorten proofs, clarify the operator package, or directly unlock a next theorem. A reference is ranked lower when it is excellent but mainly useful after the main architecture is already stable.

How to use the list

The intended workflow is:

1. Read the first five items before any major rewrite. They determine the cleanest proof architecture.
2. Read items 6–10 when preparing the bridge from the base operator to the full operator A_λ.
3. Read items 11–18 when refactoring the paper for Dedekind or automorphic portability.
4. Read items 19–23 when building the CCM comparison and harvesting finite-section or determinant ideas.
5. Read items 24–29 when sharpening constants, asymptotics, and prime-tail estimates.

The phrase *what to mine* means: extract the theorem, lemma, proof template, normalization convention, or strategy that should be imported into the paper or into a follow-up project.

1. Ranked reading list

Rank	Reference	Why it is ranked here	What to mine
1	Fukushima–Oshima–Takeda, *Dirichlet Forms and Symmetric Markov Processes*	This is the natural home of the principal jump-form piece of L_λ.	Mine the minimal package for closed symmetric forms, Friedrichs extension, Markov property, invariance criteria, irreducibility language, and any theorem that lets you replace custom operator constructions by one abstract form theorem plus local hypothesis checking.
2	Ouhabaz, *Analysis of Heat Equations on Domains*	Best single source for semigroup language tied to positivity, domination, and irreducibility.	Mine the cleanest positivity-preserving and invariance criteria, semigroup reformulations of irreducibility, and any statements that let you phrase simplicity of the ground state through positivity improvement rather than direct Euler-Lagrange analysis.
3	Keller–Lenz–Vogt–Wojciechowski, *Note on basic features of large time behaviour of heat kernels*	Gives a compact route from positivity-improving semigroups to ground-state conclusions.	Mine the abstract positivity-improving to simple-ground-state mechanism, long-time convergence statements, and the exact hypotheses under which positivity plus irreducibility force uniqueness and strict positivity of the bottom eigenfunction.
4	Kato, *Perturbation Theory for Linear Operators*	This is the main tool once the base operator is understood and the perturbation K_λ comes back in.	Mine the statements on holomorphic or continuous perturbation of isolated simple eigenvalues, spectral projections via Riesz integrals, and the templates for proving no crossing of the lowest eigenvalue in a coupling family $L_\lambda + tK_\lambda$.
5	Ben Amor, *Sobolev-Orlicz inequalities, ultracontractivity and spectra of time changed Dirichlet forms*	Best packaging reference for turning form-domain compactness into discrete spectrum.	Mine the embedding-to-compact-resolvent logic, especially any abstract formulation that avoids custom compactness arguments. The goal is to recast compact resolvent as a clean consequence of compact embedding of the energy space into L^2.
6	Arendt–Gluck on positive irreducible semigroups	Useful sharpening of the positivity spine.	Mine the exact irreducibility criteria and any principal-eigenvalue consequences that are shorter or more directly quotable than the broader semigroup references.

Rank	Reference	Why it is ranked here	What to mine
7	Arendt–ter Elst–Gluck on strict positivity of principal eigenfunctions	Focused payoff for principal-eigenfunction structure.	Mine statements that separate positivity-preserving from strict positivity, especially if they can be applied to justify that the ground state is not only nonnegative but strictly positive on the interval.
8	Keller–Lenz–Wojciechowski on symmetries and reducing subspaces	Directly relevant to the "even" part once simplicity is known.	Mine the symmetry bookkeeping: commuting involutions, decomposition into invariant sectors, and any criterion showing that uniqueness of the positive ground state forces it into the symmetric sector.
9	Gesztesy and collaborators on Birman–Schwinger methods and finite-rank perturbations	Important because K_λ is low rank.	Mine finite-dimensional determinant reductions, Birman–Schwinger reformulations of eigenvalue problems, and multiplicity bookkeeping for low-rank perturbations. These are prime candidates for ruling out bottom-eigenvalue crossings after the base theory is established.
10	Seifert–Wingert on perturbation of positive semigroups	Selective but potentially strong for deformation arguments.	Mine any theorem showing persistence of positivity or irreducibility under perturbation, especially when the perturbation is bounded, relatively bounded, or finite rank.
11	Chen–Kim–Wang, *Intrinsic Ultracontractivity and Ground State Estimates of Non-local Dirichlet forms on Unbounded Open Sets*	Best bridge to quantitative ground-state information.	Mine semigroup criteria for intrinsic ultracontractivity, ground-state upper and lower bounds, and any domination arguments that may inspire sharper estimates for the principal eigenfunction of L_λ.
12	Chen–Wang, *Intrinsic contractivity properties of Feynman-Kac semigroups for symmetric jump processes with infinite range jumps*	Useful when lower-order terms are reintroduced.	Mine techniques for adding lower-order terms to jump generators, and any contractivity or domination estimates that may survive when passing from the base operator to more general nonlocal Schrodinger type models.
13	Baraniewicz, *Ground state decay for Schrodinger operators with confining potentials*	Lower priority, but valuable for future quantitative work.	Mine the strategic ideas rather than literal hypotheses: how confining geometry plus a nonlocal kinetic term produces ground-state control, and which arguments may carry over to bounded-log-interval models.

Rank	Reference	Why it is ranked here	What to mine
14	Weil on the explicit formula	Foundational arithmetic input.	Mine the exact structural decomposition of the explicit formula into zero sum, arithmetic sum, archimedean terms, and polar contributions. The goal is to isolate which pieces should become the principal operator, which are scalar, and which are finite rank.
15	Poitou on the explicit formula for number fields	Best classical guide for Dedekind-zeta portability.	Mine the number-field archimedean decomposition, normalization conventions, and the way residual terms are organized. This is the key reference for understanding what changes and what stays stable beyond ζ.
16	Iwaniec–Kowalski, *Analytic Number Theory*	Default normalization and background source for general L-functions.	Mine completed-L notation, gamma-factor bookkeeping, functional equations, Rankin-Selberg normalizations, and any standard explicit-formula setup needed to formulate an abstract operator theorem with data supplied by a general completed L-function.
17	Buthe, *A method for proving the completeness of a list of zeros of certain L-functions*	High practical value for abstracting admissible L-function hypotheses.	Mine the general Weil-Barner explicit-formula template, especially the hypothesis list on gamma factors and Euler products. This is the right source for a clean abstract-input section in a generalized sequel paper.
18	Lagarias, *Li coefficients for automorphic L-functions*	High value for the automorphic and Weil-positivity side.	Mine the conceptual bridge between explicit formulas, Weil positivity, and automorphic L-functions. This is where to extract language showing that the operator package is not peculiar to the Riemann zeta function.
19	de la Bretèche–Kuperberg on explicit formulas for Dedekind zeta and Artin L-functions	Modern number-field explicit-formula packaging.	Mine modern normalizations, sharp statements of the explicit formula, and any presentation that makes local-archimedean pieces easier to identify as operator kernels or scalar corrections.

Rank	Reference	Why it is ranked here	What to mine
20	Battistoni–Molteni notes on explicit formulas and class-group generators	Concrete and computation-aware explicit-formula resource.	Mine formulations that are especially suited to effective estimates, computational checks, and explicit handling of number-field terms. This is useful when preparing numerical or computer-assisted versions of the theory.
21	Connes–Consani, *Spectral triples and ζ-cycles*	First CCM item to read because it gives concrete finite-section and trial-state ideas.	Mine the finite-dimensional truncation picture, Toeplitz-like matrix structure, prolate-spheroidal heuristics, and any low-rank identities that could inspire Birman–Schwinger style reductions or numerical trial vectors for the base operator.
22	Connes–Consani–Moscovici, *Zeta Spectral Triples*	Canonical comparison target for the full operator.	Mine the representation-theorem viewpoint for the full restricted Weil form, the exact place where the rank-two contribution enters, and the relation between the full canonical operator and a base-plus-low-rank decomposition. This is the central reference for showing complementarity rather than competition between the two programs.
23	Connes–van Suijlekom, *Quadratic Forms, Real Zeros and Echoes of the Spectral Action*	Best source for the entire-function payoff of simple-even.	Mine the real-zero theorem and determinant/Fourier-transform outputs that become available once simple-even is proved. This turns the spectral theorem into a visibly stronger consequence than ground-state uniqueness alone.
24	Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*	Valuable as a strategic map of what remains open in Connes' program.	Mine the exact formulation of Step 1 and Step 2, the role of the approximation problem, and any remarks on prolate-wave approximation or numerical evidence that can be used to orient the long-term plan for A_λ.
25	Titchmarsh–Heath-Brown, *The Theory of the Riemann Zeta-Function*	Core source for classical zeta analysis and zero-counting formulas.	Mine Riemann–von Mangoldt, explicit zero-counting formulas, Hadamard product facts, and classical asymptotics that can calibrate spectral windows, determinant comparisons, or trace-style arguments.

Rank	Reference	Why it is ranked here	What to mine
26	Edwards, *Riemann's Zeta Function*	Cleaner conceptual companion to Titchmarsh for some formulas.	Mine alternative derivations and normalizations of the functional equation, Ξ-function factorization, and classical product formulas that may simplify the archimedean or determinant side of the operator analysis.
27	Montgomery–Vaughan, *Multiplicative Number Theory I: Classical Theory*	Best source for prime-sum technology behind quantitative estimates.	Mine prime-number-theoretic estimates, summation tools, and average-order facts needed to control prime tails, compare truncated and full arithmetic sums, and turn qualitative explicit-formula decompositions into effective inequalities.
28	DLMF Chapter 5 on the gamma function	Essential reference for archimedean simplification.	Mine gamma reflection, duplication, multiplication, digamma identities, and Stirling-type asymptotics. These should be used to simplify the scalar term C_λ, identify cancellations, and derive large-λ asymptotics cleanly.
29	Rosser–Schoenfeld and Dusart on explicit prime estimates	Main source for effective tail bounds.	Mine explicit inequalities for $\pi(x)$, $\vartheta(x)$, $\psi(x)$, prime sums, and logarithmic sums. These are the references that should feed directly into quantitative truncation bounds, numerical implementations, and explicit error terms in derived inequalities for L_λ.

2. Immediate extraction plan

If the goal is to improve the current paper before broadening scope, the first-pass extraction order should be:

1. **Fukushima–Oshima–Takeda + Ouhabaz.** Rewrite the paper so that the principal jump form is treated by one abstract theorem on closed symmetric Markov forms and positive semigroups.
2. **Keller–Lenz–Vogt–Wojciechowski.** Replace hand-crafted simplicity arguments with a positivity-improving route whenever the hypotheses can be verified.
3. **Ben Amor.** Repackage compact resolvent as compact embedding of the form domain into L^2.
4. **Kato + Gesztesy-style Birman–Schwinger sources.** Build the perturbative bridge from the base operator to A_λ.

5. **Weil + Poitou + Iwaniec–Kowalski + Buthe.** Separate the abstract operator theorem from the arithmetic verification stage so the paper becomes extensible.

If the goal shifts specifically to simple-even for A_λ, the shortest high-yield reading chain is:

Fukushima–Oshima–Takeda → Ouhabaz → Keller–Lenz–Vogt–Wojciechowski → Kato → Birman–Schwinger / fin

If the goal shifts to stronger estimates, then after the operator spine one should prioritize:

DLMF → Montgomery–Vaughan → Rosser–Schoenfeld / Dusart → Titchmarsh / Edwards.

3. Reading by goal

3.1. For rewriting and simplifying the current paper

Read items 1–5 first. These are the references most likely to shorten proofs, standardize statements, and remove custom arguments.

3.2. For Dedekind zeta, automorphic, and Rankin-Selberg extensions

Read items 15–20 in parallel with item 16. The key output is an abstract input package: completed L-function, gamma data, explicit formula, and classification of remainder terms by operator type.

3.3. For stronger spectral conclusions and perturbative control

Read items 3, 4, 6, 7, 8, 9, and 10. Together these are the best route to simple ground state, symmetry-sector control, and no-crossing strategies.

3.4. For new structural properties of L_λ

Read items 11–13 and 25–29 selectively. These will not be the shortest route to the first main theorem, but they are the best source of ideas for ground-state bounds, asymptotics, and effective inequalities.

3.5. For cross-fertilization with the CCM program

Read items 21–24 after the base operator architecture is stable. Their main value is canonicality language, finite-section heuristics, determinant outputs, and trial-vector ideas.

Bottom line

The highest-yield reading sequence is not broad. It begins with the Dirichlet-form and positive-semigroup spine, then moves to perturbation theory, then to explicit-formula portability, and only

after that to CCM refinements and quantitative number-theoretic sharpening. The practical lesson is simple: read first for *proof architecture*, then for *generalization*, then for *numerical and quantitative sharpening*.